Approach: same data, same stop rule, GD vs L-BFGS

DATASET	GD UPDATES	L-BFGS ITERATIONS	GD GRADIENT EVALS	L-BFGS GRADIENT EVALS
κ = 1	1	2	2	3
κ = 10	75	6	76	7
κ = 100	**746**	**5**	747	10

746 → 5

iterations at κ = 100, GD vs L-BFGS (preliminary)

GD matches theory exactly: predicted 1, 75, 746

HOW WE RUN IT

Start w = (0, 0), stop when gradient norm < 1e-6. GD uses step 2/(λmin+λmax). L-BFGS keeps the last 10 gradient changes and uses a line search. Both are counted the same way.

LIMITS

Noise-free, 2 parameters, full-batch gradients. Says nothing yet about noisy minibatches, where L-BFGS is known to be unstable.

Our own runs, preliminary. Gradient evaluations count distinct points. Cross-checked against scipy L-BFGS-B (2, 6, 5 iterations). Setup from the project README.